DISJOINT CLOSED SUBSETS ARE SEPARATED BY A GLOBAL FUNCTION IN THE VANISHING SITUATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The reduced induced subspace structure on a closed subset. Properties of Spaces, definition spaces-properties-definition-reduced-induced-space (invoked at proof lines 3080-3081). It endows Z_0 and Z_1 with the reduced structure so that the inclusions i_0 and i_1 are closed immersions of algebraic spaces.
- Vanishing of H^1 on the kernel of the comparison map in the vanishing situation. Same chapter spaces-cohomology.tex, Situation \ref{situation-vanishing} and the vanishing of H^1 for quasi-coherent modules there (used at proof line 3090). It makes the map on global sections induced by a surjection of quasi-coherent modules surjective, which is what produces the element $a \in A$.

Source reference. Original reference 07UY: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2991–2991. Navigation pointer for \ref{situation-vanishing} used inside the counted span (line 3073); the argument that consumes it is retained in the span.

Source reference. Original definition 047X: sources/stacks-chow/tags.txt, source lines 5524–5524. Navigation pointer for \ref{spaces-properties-definition-reduced-induced-space} used inside the counted span (line 3081); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 07UY. Let $Z_0, Z_1 \subset |X|$ be disjoint closed subsets. Then there exists an $a \in A$ such that $Z_0 \subset V(a)$ and $Z_1 \subset V(a-1)$.*

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Proof. We may and do endow Z_0, Z_1 with the reduced induced subspace structure (Properties of Spaces, Definition 047X) and we denote $i_0 : Z_0 \to X$ and $i_1 : Z_1 \to X$ the corresponding closed immersions. Since $Z_0 \cap Z_1 = \emptyset$ we see that the canonical map of quasi-coherent $\mathcal{O}_X$-modules

$$\mathcal{O}_X \longrightarrow i_{0,*}\mathcal{O}_{Z_0} \oplus i_{1,*}\mathcal{O}_{Z_1}$$

is surjective (look at stalks at geometric points). Since $H^1(X, -)$ is zero on the kernel of this map the induced map of global sections is surjective. Thus we can find $a \in A$ which maps to the global section $(0, 1)$ of the right hand side. $\square$